2024 期中高数上试题（回忆版）

1. Multiple Choice Questions: (only one correct answer for each of the following questions.)

(1) Suppose that $\int_1^5 f(x)dx=3$, $\int_5^6 f(x)dx=2$, $\int_1^6 g(x)dx=2$.

Then $\int_1^6 (f(x)-2g(x))dx=$

(A) -1 (B) 1 (C) 3 (D) None of (A), (B) and (C) is correct.

(2) How many real roots does the equation $x^3=2x^2+3x-3$ have?

(A) 0 (B) 1 (C) 2 (D) 3

(3) If $f(x)$ is twice differentiable on $[0,1]$ and $f''(x)>0$, then which of the following statements is correct?

(A) $f'(1)>f'(0)>f(1)-f(0)$ (B) $f'(1)>f(1)-f(0)>f'(0)$

(C) $f(1)-f(0)>f'(1)>f'(0)$ (D) $f'(0)>f(1)-f(0)>f'(1)$

(4) Let $f(x)=\begin{cases} x^2\sin\frac{1}{x}, & x\neq 0 \\ 0, & x=0 \end{cases}$. Which of the following statements is correct?

(A) $f''(0)=0$ and $(0,0)$ is a point of inflection.

(B) $f''(0)=0$ and $(0,0)$ is not a point of inflection.

(C) $f''(0)$ does not exist and $(0,0)$ is a point of inflection.

(D) $f''(0)$ does not exist and $(0,0)$ is not a point of inflection.

(5) Which of the following statements must be correct?

(A) If $f(x)$ is differentiable at $x=a$, then $|f(x)|$ is differentiable at $x=a$.

(B) If $|f(x)|$ is differentiable at $x=a$, then $f(x)$ is differentiable at $x=a$.

(C) If $f(x)$ is differentiable at $x=a$ and $f(a)=0$, $f'(a)=0$, then $|f(x)|$ is differentiable at $x=a$.

(D) If $f(x)$ is differentible at $x=a$ and $f(a)=0$, $f'(a)\neq 0$, then $|f(x)|$ is differentiable at $x=a$.

2. Fill in the blanks.

(1) Let $f(x)=\begin{cases} \sqrt{1-(x-1)^2}, & \text{if } 0\leq x\leq 2 \\ -\sqrt{4-(x-4)^2}, & \text{if } 2\leq x\leq 6 \end{cases}$

According to the relationship between definite integral and area, $\int_0^6 f(x)dx=$ ____.

(2) The linearization of $f(x)=\sqrt[3]{1+5x^4}$ at $x=0$ is $L(x)=$ _____.

(3) Let $f(x)=x^3+3x+1$. Use Newton's method to find the root of $f(x)=0$. Start with $x_0=1$, then $x_2=$ _____.

(4) If $f(x)+x\sin(f(x))=x^2+1$, then $f'(0)=$ _____.

(5) If $\lim_{x\to\infty} f'(x)=10$, then $\lim_{x\to\infty}(f(x+10)-f(x))=$ _____.

3. A rectangle is to be inscribed in the ellipse $\frac{x^2}{4}+y^2=1$.

What should the dimensions of the rectangle be to maximize its area? What is the maximum area?

4. Let $\frac{d}{dx}f(\sin x)|_{x=0} = \frac{d}{dx}f^2(\sin x)|_{x=0}$,
and $f'(0) \neq 0$. Find $f(0)$.

5. Determine if the following limits exist or not. If so, find the limit. If not, explain why.
(L'Hopital's Rule is not allowed to be used.)

(1) $\lim_{x\to 1} \frac{x^4-1}{\sqrt{3+x}-2}$

(2) $\lim_{x\to -\infty} \frac{x^2 \sin\frac{1}{2x}}{\sqrt{x^2+2024x+1}}$

6. Let $f(x) = x^{\frac{2}{3}}(6-x)^{\frac{1}{3}}$.

(1) Indentify where the local extrema of f occur. Find the function's local extreme values.

(2) Find the open intervals where the graph of f is concave up and where it is concave down.

(3) Sketch the graph.

7. Find $a \neq 0$ such that the curves $y = \frac{1}{2}x^2$ and $y^2 + xy = a$ are tangent to each other, and find the equation of the tangent line at the point of tangency.

8. Assume that f is continuous on $[0,\infty)$, differentiable on $(0,\infty)$, $f(0)=0$ and f' is increasing on $(0,\infty)$. Prove that $\frac{f(x)}{x}$ is also increasing on $(0,\infty)$.